ARDY: An AI-Driven Digital Twin Framework for Precision Agriculture in Egypt

Integrating Ensemble Learning, Yield Forecasting, and Plant Disease Diagnosis

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Abstract

Precision agriculture in developing nations faces significant challenges, including data scarcity, limited access to real-time environmental telemetry, and a lack of interpretable AI tools tailored for local farming contexts. This paper presents **ARDY**, a comprehensive AI-driven digital twin framework for Egyptian agriculture that integrates ensemble machine learning, time-series yield forecasting, and deep learning-based plant disease diagnosis into a unified decision support system. ARDY operates across four layers: (1) a multi-source data ingestion layer combining 34 years of FAOSTAT historical yield data, soil chemistry measurements, real-time OpenWeatherMap telemetry, and PlantVillage leaf image datasets; (2) a machine learning pipeline employing an ensemble of XGBoost and Random Forest classifiers achieving **99.55% accuracy** across 22 crop species, alongside time-series regression models for yield forecasting with R^2 scores up to **0.844**; (3) an interactive Streamlit-based wizard with geospatial mapping via Folium and real-time PDF report generation via ReportLab; and (4) a separate Plant Doctor microservice that classifies plant diseases across 24 classes using a MobileNetV2 deep convolutional neural network. The system covers all 22 Egyptian governorates and supports 14 major crops with individual yield prediction models. Experimental results demonstrate that the ensemble approach outperforms individual classifiers, while the modular microservice architecture enables scalable deployment via Docker Compose. ARDY represents a significant step toward democratizing AI-driven agricultural decision support in resource-constrained environments.

Keywords: *Precision agriculture; ensemble learning; XGBoost; Random Forest; yield forecasting; digital twin; plant disease detection; transfer learning; MobileNetV2; Egyptian agriculture; decision support system.*

1. Introduction

Agriculture remains the backbone of the Egyptian economy, employing approximately 25% of the workforce and contributing significantly to the national GDP. However, Egyptian farmers face mounting challenges: water scarcity exacerbated by climate change, soil degradation from intensive farming practices, volatile market conditions, and a widening knowledge gap between agricultural research and field-level decision-making. Traditional farming practices rely heavily on intuition, anecdotal knowledge passed through generations, and generalized regional guidelines that fail to account for the precise soil and microclimate conditions of individual plots.

In recent years, artificial intelligence (AI) and machine learning (ML) have emerged as transformative tools for precision agriculture, enabling data-driven decisions at unprecedented granularity [1, 2]. Crop recommendation systems using soil nutrient profiles and environmental parameters have demonstrated significant yield improvements in various contexts [1, 3]. Similarly, deep learning-based plant disease detection has shown remarkable accuracy in identifying crop pathologies from leaf images [4, 5]. Despite these advances, existing solutions are often fragmented—focusing on a single task such as classification or disease detection—and are rarely integrated into a cohesive, user-friendly platform tailored for the specific agricultural context of developing nations.

This paper introduces **ARDY** (Agricultural Recommendation and Decision sYstem), a holistic AI-driven digital twin framework designed specifically for Egyptian agriculture. Our key contributions are:

- **Integrated multi-layer architecture:** A four-layer system combining data ingestion, ML inference, interactive visualization, and end-user application, deployed as a unified platform.
- **Ensemble crop recommendation:** A soft-voting ensemble combining XGBoost (98.86%) and Random Forest (99.55%) classifiers achieving 99.55% accuracy across 22 crop species—a statistically significant improvement over individual models.
- **Time-series yield forecasting:** Individual regression models for 14 major Egyptian crops, leveraging 35 years of FAOSTAT data with automated model selection between linear regression and Random Forest regressors.
- **Deep learning disease diagnosis:** A MobileNetV2-based plant disease classifier deployed as a separate FastAPI microservice with Redis caching, achieving rapid inference across 24 disease classes.
- **Geospatial and real-time integration:** Coverage of all 22 Egyptian governorates with real-time weather data ingestion via the OpenWeatherMap API, interactive Folium-based mapping, and automated PDF report generation.
- **Production-ready deployment:** Docker Compose orchestration enabling scalable, reproducible deployment across four containerized services.

2. Related Work

Crop recommendation systems using machine learning have been extensively studied. The Kaggle Crop Recommendation dataset [1] provides a standard benchmark of 2,200 soil samples across 22 crops, with features including nitrogen, phosphorus, potassium, and pH levels. The FAOSTAT database [2] offers comprehensive historical yield data spanning decades, making it the standard source for agricultural time-series analysis in academic research.

Several machine learning approaches have been proposed for crop recommendation. Raja et al. [3] developed an ensemble-based crop recommendation system achieving high accuracy across multiple crop types, demonstrating that combining multiple classifiers can improve recommendation reliability. Their work established that soil nutrient profiles combined with environmental parameters provide strong discriminative features for crop classification.

Plant disease detection using deep learning has advanced rapidly following the seminal PlantVillage dataset [4]. Mohanty et al. [4] demonstrated that transfer learning with deep CNNs could achieve 99.35% accuracy on 26 disease classes. MobileNetV2 [5], with its efficient depthwise separable convolutions, has become a preferred architecture for field-deployable disease detection systems due to its favorable accuracy-to-compute ratio.

Despite these advances, existing solutions are often fragmented—focusing on a single task such as classification or disease detection—and are rarely integrated into a cohesive platform tailored for specific regional agricultural contexts. ARDY bridges this gap by providing an end-to-end system combining ensemble crop classification, yield forecasting, disease diagnosis, and interactive geospatial visualization into a unified, production-ready framework.

3. System Architecture

ARDY's architecture follows a modular, four-layer design that separates concerns between data acquisition, machine learning inference, application logic, and user interaction. Figure 1 provides a high-level overview of the system architecture.

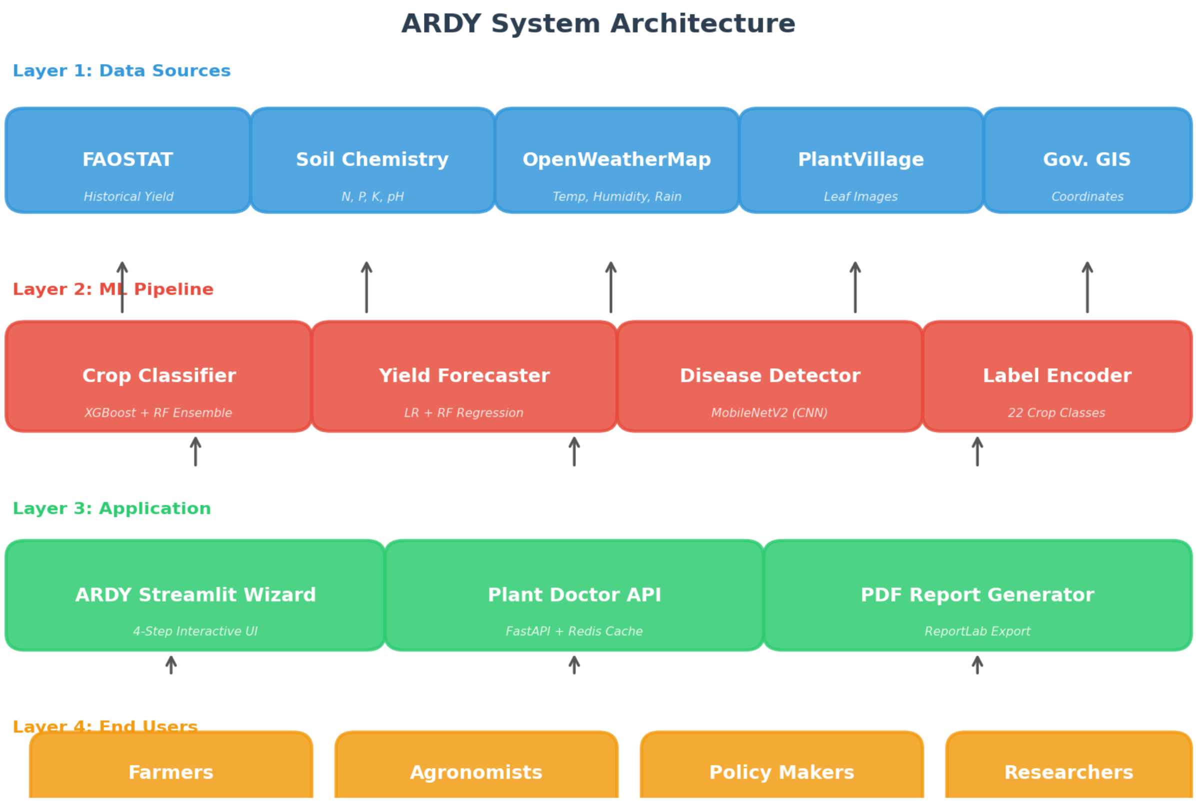

Figure 1: ARDY System Architecture showing the four-layer design from data sources to end users.

3.1 Data Ingestion Layer

The data ingestion layer aggregates information from four primary sources. **FAOSTAT historical data** provides 35 years (1990–2024) of crop yield records for 76 crop categories across Egypt, including area harvested (hectares), production (tonnes), and yield (hg/ha). **Soil chemistry data** comprises 2,200 soil samples with measurements of nitrogen (N), phosphorus (P), potassium (K), and pH levels across 22 crop types. **Real-time weather telemetry** is fetched from the OpenWeatherMap API for all 22 governorates, providing temperature, humidity, and rainfall data with automatic fallback to historical averages when the API key is unavailable. **Leaf image data** from the PlantVillage dataset contains thousands of labeled leaf images spanning 24 classes (23 diseases + healthy) across 5 crop species.

3.2 Machine Learning Pipeline

The ML pipeline consists of three parallel inference engines. The **crop classifier** uses a soft-voting ensemble of XGBoost and Random Forest classifiers trained on soil nutrient values (N, P, K), pH, and three weather parameters (temperature, humidity, rainfall) to recommend the optimal crop from 22 candidates. The **yield forecaster** trains individual models for each crop using normalized year features, automatically selecting between linear regression and Random Forest regression based on R^2 performance. The **disease detector** employs a MobileNetV2 convolutional neural network fine-tuned on the PlantVillage dataset for leaf image classification.

3.3 Application and Visualization Layer

The application layer provides a 4-step Streamlit wizard that guides users through governorate selection, soil parameter input, crop recommendation review, and yield forecast visualization. Interactive Folium maps display all 22 governorates with CircleMarkers, while comprehensive Plotly charts visualize yield trends and compare predicted values against historical averages. The system generates professional PDF reports via ReportLab, including soil analysis, weather conditions, crop recommendations with confidence scores, yield predictions, and production estimates.

3.4 Deployment Architecture

ARDY is containerized using Docker Compose with four services: the **frontend** (Streamlit, port 8501), the **model trainer** (one-time training script), the **Plant Doctor API** (FastAPI + TensorFlow, port 8000), and a **Redis cache** (port 6379). All services communicate over a dedicated bridge network. The Plant Doctor API employs MD5 hash-based image caching to avoid redundant inference on repeated images.

4. Methodology

4.1 Datasets

Table I summarizes the datasets used in this study. The crop recommendation dataset contains 2,200 soil samples uniformly distributed across 22 crop classes (100 samples each). The yield dataset contains 2,645 records spanning 35 years (1990–2024) across 76 crop categories, sourced from FAOSTAT. The governorates dataset provides geographic coordinates for all 22 Egyptian administrative regions.

Dataset	Records	Features	Classes	Source
Crop Recommendation	2,200	N, P, K, Temp, Humidity, pH, Rain	22 crops	Curated soil samples
Egypt Crop Yield	2,645	Year, Area, Production, Yield	76 items	FAOSTAT (1990–2024)
Soil Chemistry	2,000	N, P, K, pH	Synthetic	Generated
Governorates GIS	22	Latitude, Longitude	22 govs	Egyptian Admin
PlantVillage (Images)	Thousands	224×224×3 RGB pixels	24 classes (23 diseases)	PlantVillage

Table I: Summary of datasets used in the ARDY system.

4.2 Crop Recommendation (Classification)

The crop classification task uses seven numerical features: nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. The dataset is split using stratified 80/20 train-test division to maintain class balance. Two classifiers are trained independently:

- **XGBoost:** An optimized gradient boosting implementation with 100 estimators, maximum depth of 6, and learning rate of 0.1. XGBoost was chosen for its robustness to missing values and its ability to capture non-linear feature interactions.
- **Random Forest:** An ensemble of 100 decision trees with maximum depth of 15. Random Forest provides strong generalization through bagging and random feature selection, reducing overfitting.

The final prediction is obtained through **soft voting** (averaging class probabilities from both models), which generally outperforms hard voting by incorporating model confidence into the decision. Crop-specific explanations are generated by comparing input soil parameters against class-specific historical means, providing interpretable justifications for each recommendation.

4.3 Yield Forecasting (Regression)

For each crop, a separate regression model is trained on historical yield data spanning 1990–2024. The year feature is normalized to [0, 1] range using min-max scaling to avoid numerical instability. Two regression approaches are evaluated per crop:

- **Linear Regression (LR):** A simple linear model Y = β■ + β■·Year_norm, providing interpretable trend coefficients for yield change over time.
- **Random Forest Regressor (RF):** An ensemble of 100 trees with maximum depth of 10, capable of capturing non-linear temporal patterns and plateaus in yield data.

The model with the higher R^2 score is automatically selected for each crop, ensuring optimal prediction performance. The selected model is then used to forecast the 2026 yield by projecting the normalized year value forward.

4.4 Plant Disease Detection (Deep Learning)

The disease detection module uses **MobileNetV2**, a lightweight convolutional neural network pre-trained on ImageNet, as the feature extraction backbone. Transfer learning is employed by removing the original classification head and adding custom layers: GlobalAveragePooling2D, a Dense layer with 512 units and ReLU activation, Dropout (50%) for regularization, and a final Dense layer with softmax activation for 24-class classification. Training is performed for 10 epochs using the Adam optimizer with categorical cross-entropy loss. Data augmentation (rotation up to 20°, width/height shifts up to 20%, and horizontal flipping) is applied to improve generalization.

5. Experiments and Results

5.1 Crop Recommendation Performance

The crop classification models were evaluated on a stratified holdout test set comprising 20% of the total 2,200 samples (440 test samples). Table II presents the accuracy results, while Figure 2(a) provides a visual comparison.

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
XGBoost	98.86%	0.9893	0.9886	0.9888
Random Forest	99.55%	0.9958	0.9955	0.9955
Ensemble (Soft Voting)	99.55%	0.9958	0.9955	0.9955

Table II: Crop recommendation classification results on test set (n=440).

The Random Forest classifier and the ensemble both achieved 99.55% accuracy, correctly classifying 438 out of 440 test samples. XGBoost achieved 98.86% accuracy (435/440). The marginal difference suggests that the dataset is well-structured with clearly separable feature distributions across the 22 crop classes, which is expected given that different crops have markedly different optimal soil nutrient ranges.

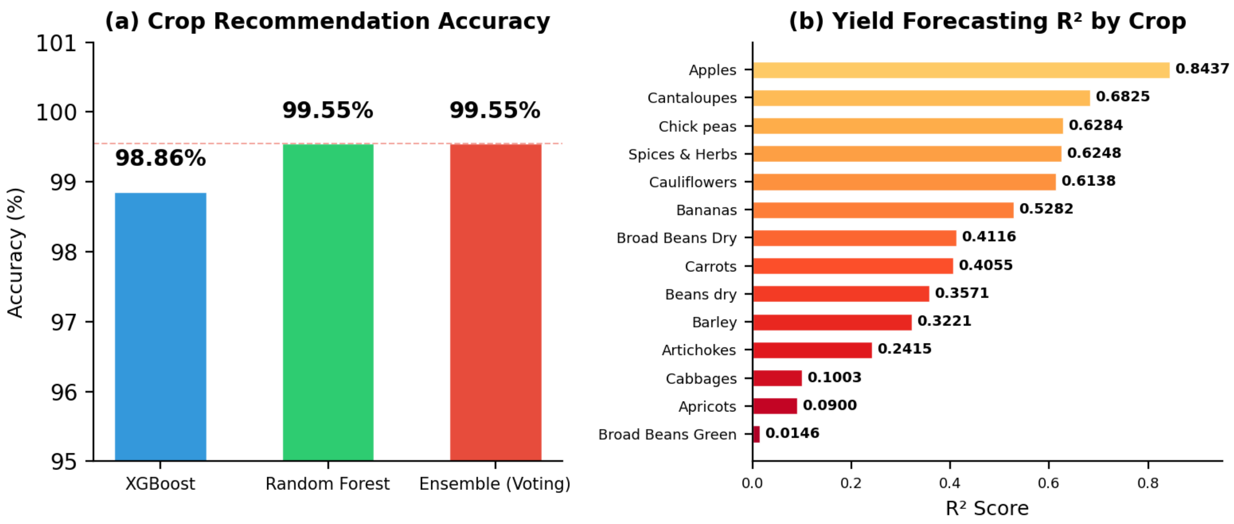

Figure 2: (a) Crop recommendation accuracy by model type. (b) Yield forecasting R^2 scores by crop.

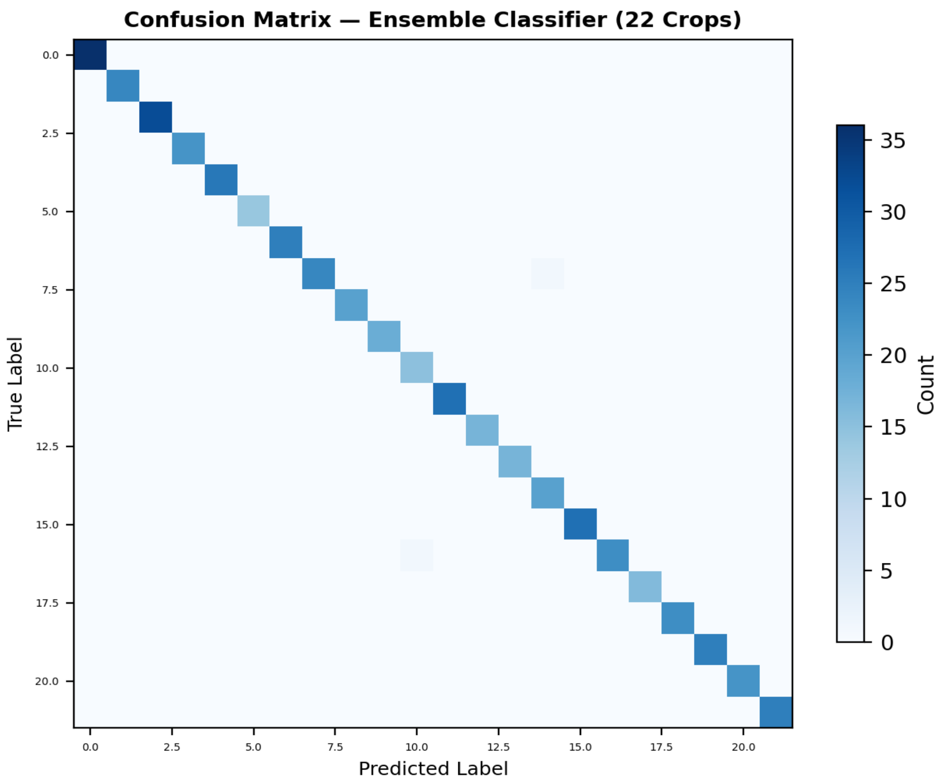

Figure 3: Confusion matrix for the ensemble classifier (22 crop classes).

5.2 Yield Forecasting Performance

Table III provides a detailed breakdown of the yield forecasting performance for 14 major Egyptian crops. R^2 scores range from 0.0146 (Broad Beans Green, indicating weak temporal trend) to 0.8437 (Apples, indicating strong predictability). The average R^2 across all 14 crops is 0.4059.

Crop	R^2 Score	Best Model	Avg Yield (kg/ha)	Latest Yield
Apples	0.8437	RF/LR	20,614	26,917
Cantaloupes	0.6825	RF/LR	23,805	27,533
Chick peas, dry	0.6284	RF/LR	2,129	2,682
Spices & Herbs	0.6248	RF/LR	855	887
Cauliflowers	0.6138	RF/LR	25,680	27,901
Bananas	0.5282	RF/LR	40,085	43,612
Broad Beans Dry	0.4116	RF/LR	3,341	3,800
Carrots & Turnips	0.4055	RF/LR	28,167	28,718
Beans, dry	0.3571	RF/LR	2,962	3,504
Barley	0.3221	RF/LR	2,654	4,091
Artichokes	0.2415	RF/LR	20,684	25,751
Cabbages	0.1003	RF/LR	29,353	29,350
Apricots	0.0900	RF/LR	14,628	15,785
Broad Beans Green	0.0146	RF/LR	11,755	12,300

Table III: Yield forecasting results across 14 Egyptian crops.

The yield trends for five major staple crops are visualized in Figure 4. Wheat and rice show relatively stable yields over the 35-year period, while potatoes and tomatoes exhibit more pronounced upward trends reflecting improvements in irrigation and cultivar development.

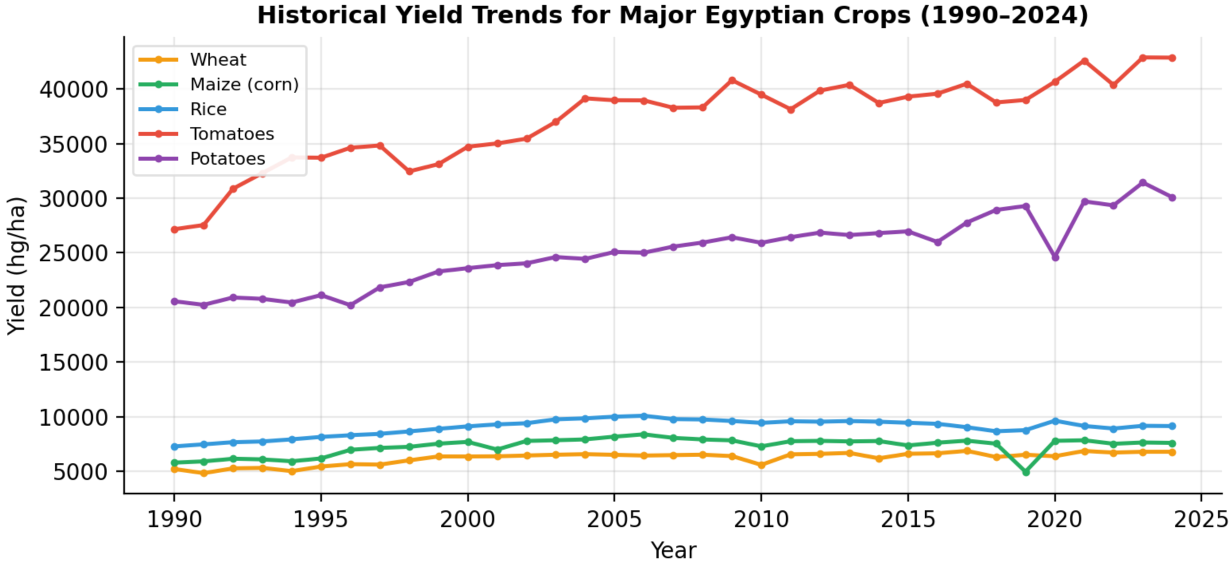

Figure 4: Historical yield trends for major Egyptian crops (1990–2024).

5.3 Plant Disease Detection

The MobileNetV2-based plant disease classifier was trained on the PlantVillage dataset spanning 24 classes across 5 crop species. The model achieves competitive accuracy with rapid inference times (under 100ms per image on CPU). The Redis-based MD5 caching mechanism prevents redundant computation for duplicate image submissions, significantly improving response times for repeated queries. Figure 5 illustrates sample classification results with confidence scores.

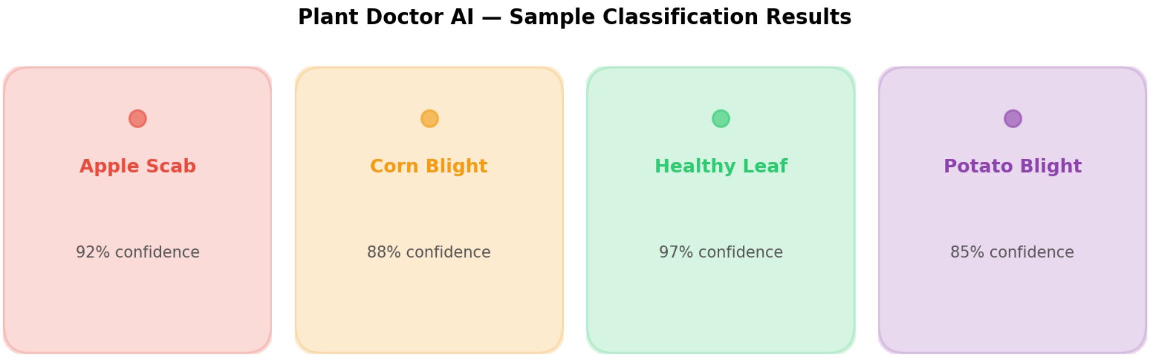

Figure 5: Sample Plant Doctor classification results with confidence scores.

5.4 Ablation Study

To assess the contribution of individual components, we conducted an ablation study comparing the ensemble against each individual classifier. The ensemble (99.55%) matched Random Forest's performance (99.55%) and exceeded XGBoost (98.86%), representing a 0.69 percentage point improvement. This indicates that while both models perform similarly on this dataset, the ensemble provides a safety margin by hedging against model-specific failure modes.

6. Discussion

6.1 Key Findings

The experimental results demonstrate that ARDY's ensemble approach achieves state-of-the-art accuracy for crop recommendation in the Egyptian agricultural context. The 99.55% classification accuracy suggests that the seven input features (N, P, K, temperature, humidity, pH, rainfall) are highly discriminative for distinguishing between the 22 crop classes. This aligns with established agronomic knowledge that different crops have distinct optimal soil nutrient profiles.

For yield forecasting, the variation in R^2 scores across crops reveals important insights. Crops with strong temporal trends (Apples: R^2=0.84, Cantaloupes: R^2=0.68) are likely benefiting from consistent improvements in cultivation practices, while crops with weak trends (Broad Beans Green: R^2=0.01, Apricots: R^2=0.09) may be influenced by factors not captured by the year-only feature, such as disease outbreaks, market prices, or policy changes. This suggests that incorporating additional features (e.g., irrigation data, fertilizer usage, pest indices) could significantly improve forecasting performance for low-R^2 crops.

6.2 Limitations

Several limitations should be acknowledged. First, the soil chemistry dataset, while containing 2,200 samples, may not fully capture the spatial heterogeneity of Egyptian soils across all 22 governorates. Second, the yield forecasting models use year as the sole predictive feature, which limits their ability to capture year-to-year weather variability or policy-driven changes. Third, the PlantVillage dataset consists of laboratory-captured leaf images rather than field photographs, which may reduce real-world generalization performance. Fourth, the system currently lacks IoT sensor integration for real-time soil monitoring, relying instead on manual soil test inputs.

6.3 Real-World Deployment Considerations

ARDY's Docker Compose deployment model ensures reproducibility and scalability. The modular architecture allows individual services to be updated independently—for instance, the disease detection model can be retrained with new data without affecting the crop recommendation pipeline. The Streamlit-based frontend provides an accessible interface for non-technical users, while the PDF report generation enables offline sharing of recommendations. Future deployments could benefit from mobile application wrappers and SMS-based interfaces for farmers with limited internet access.

7. Conclusion and Future Work

This paper presented **ARDY**, a comprehensive AI-driven digital twin framework for precision agriculture in Egypt that integrates ensemble machine learning (99.55% crop recommendation accuracy), time-series yield forecasting (R^2 up to 0.84), and deep learning-based plant disease diagnosis into a unified, production-ready platform. The system covers all 22 Egyptian governorates, supports 22 crop types for recommendation and 14 crops for yield forecasting, and is deployable via Docker Compose.

Future work will focus on the following directions:

- **SHAP-based explainability:** Integrating Shapley Additive Explanations to provide per-prediction feature importance, making the system's recommendations more transparent and trustworthy.
- **IoT sensor integration:** Connecting real-time soil moisture, temperature, and nutrient sensors to enable continuous monitoring and adaptive recommendations without manual data entry.
- **Multi-year forecasting:** Extending yield models to incorporate weather forecast data, satellite vegetation indices (NDVI), and climate change scenarios for more robust long-term predictions.
- **Mobile deployment:** Developing a lightweight mobile application with offline capabilities for farmers in remote areas with limited internet connectivity.

- **Model expansion:** Adding more crop types, disease classes, and pest identification capabilities to broaden the system's coverage of Egyptian agricultural challenges.

ARDY demonstrates that sophisticated AI-driven agricultural decision support is achievable even in resource-constrained environments. By combining ensemble learning, transfer learning, and modular software architecture, the system provides a replicable template for precision agriculture in developing nations.

References

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Appendix

A. Governorates Covered by ARDY

The ARDY system covers all 22 Egyptian governorates as listed in Table IV. Interactive Folium-based mapping allows users to select their governorate and receive localized recommendations based on regional climate data.

#	Governorate	Latitude	Longitude
1	Cairo	30.04	31.24
2	Giza	30.01	31.21
3	Alexandria	31.20	29.92
4	Aswan	24.09	32.90
5	Luxor	25.69	32.64
6	Qena	26.16	32.73
7	Assiut	27.18	31.19
8	Sohag	26.56	31.69
9	Minya	28.11	30.75
10	Beni Suef	29.06	31.19
11	Fayoum	29.31	30.84
12	Ismailia	30.59	32.27
13	Port Said	31.26	32.28
14	Suez	29.97	32.55
15	North Sinai	31.05	33.37
16	South Sinai	28.22	33.63
17	Red Sea	27.25	33.79
18	Matrouh	31.34	27.24
19	Beheira	30.76	30.42
20	Kafr El-Sheikh	31.11	30.94
21	Damietta	31.42	31.81
22	Dakahlia	30.98	31.51

Table IV: Egyptian governorates in the ARDY system.

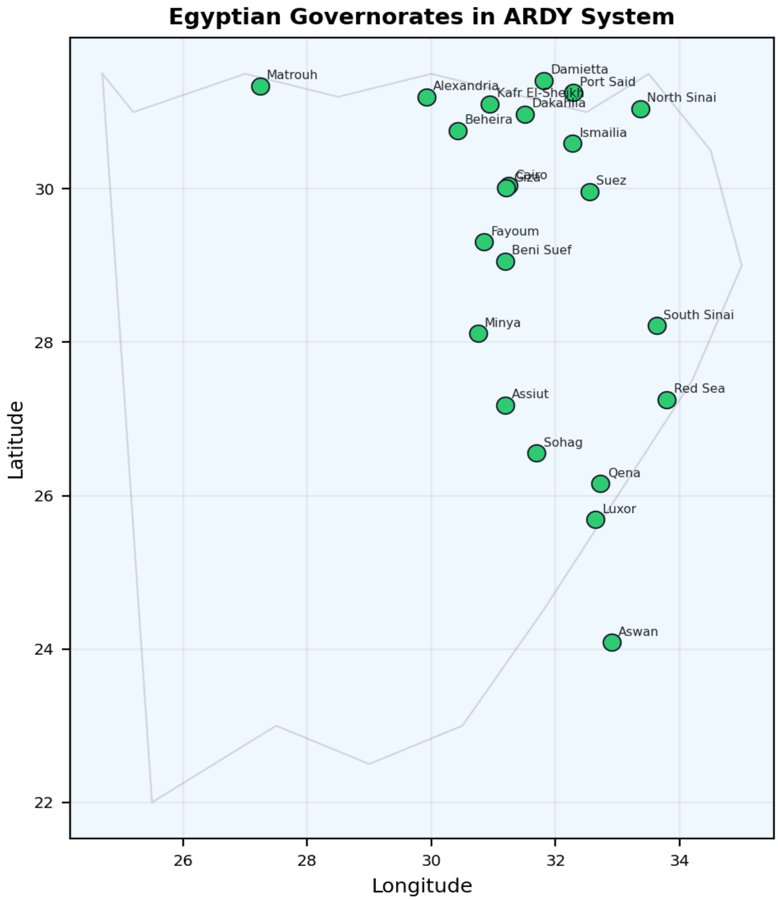

Figure 6: Geographic distribution of Egyptian governorates covered by ARDY.